

# Argyrios Gerogiannis

✉ argyrger@gmail.com | 🏠 argyrger.com | 📧 Grimoid | 📺 argyrger | 🎓 Argyrios Gerogiannis

## Research Interests

Reinforcement Learning (RL), Reasoning in LLMs and Multimodal Models, Post-Training Methods, Transformer Theory

## Education

### PhD in Electrical and Computer Engineering

University of Illinois at Urbana-Champaign (UIUC), USA

**Cumulative GPA: 4.00/4.00**

Aug. 2022 - May. 2027 (Expected)

### Diploma (BSc+MSc) in Electrical and Computer Engineering (Rank: 1/305)

University of Patras (UoP), Greece

**Cumulative GPA: 9.69/10.00**

Sep. 2017 - Aug. 2022

## Work Experience

### Uber Technologies, Inc: Machine Learning Engineer Intern

Summer 2026 / Sunnyvale, CA

- Delivered **two production-facing workstreams** within Uber Risk, each improving an existing production pipeline.
- Developed a transfer-learning approach for fraud detection, improving over production by **44% on Rides** and **10% on Eats**.
- Designed an RL framework for fraud treatment selection, yielding a projected **\$10M+ improvement** over production.

### Netradyne, Inc: AI Research Intern

Summer 2025 / San Diego, CA

- Developed a novel end-to-end autonomous driving **vision-encoder RL** stack with **DINOv2**-based vision transformers.
- Achieved **state-of-the-art performance** with **10%** gain over strong baselines on standard driving benchmarks.
- Scaled training via a **distributed** multi-simulator, parallel pipeline across **multi-GPU AWS clusters**, enabling **20x speedup**.

## Selected Research Experience

### Multimodal Model Post-Training

2026 - Present

- Developed a **low-cost** retroactive tree-based RL method **without online branching** for efficient post-training.
- Improved over state-of-the-art GRPO baselines in Speech LLMs, with up to **39%** performance gains, with **95%** lower training cost.
- Validated across **4 models variants**: Granite 4.0 1B Speech, Granite Speech 3.3 2B & 8B, Qwen-2 Audio 7B.

### Large Language Model Reasoning

2025 - Present

- Developed an **MCTS-style framework** for optimal breadth-depth allocation in LLM reasoning under compute budgets.
- Improved accuracy by **+18.8% pass@N** via intrinsic depth refinement on aligned models vs. standard i.i.d. sampling.
- Validated across **8 model variants**: OLMo-3 7B, Qwen-3 4B (Base, Instruct, RLVR) and Nemotron-3 30B (Base, Instruct).

### Non-Stationary Reinforcement Learning

2023 - Present

- Derived novel **minimax guarantees** for non-stationary RL and built the first algorithm with matching **optimal bounds**.
- Designed the first **practical black-box** for non-stationary bandits, with **improved order-optimal** performance bounds.
- Attained up to **8x** performance gains and up to **50%** lower compute vs. state-of-the-art baselines in real-world benchmarks.

## Publications

*DAL: A Practical Prior-Free Black-Box Framework for Piecewise Stationary Bandits*

**Argyrios Gerogiannis**, Yu-Han Huang, Subhonmesh Bose, Venugopal V. Veeravalli

**International Conference on Machine Learning - ICML 2026**



*LEAF: Growing Trees Without Branching for Speech-Aware Large Language Model Post-Training*

**Argyrios Gerogiannis**, Yekaterina Yegorova, Mark Hasegawa-Johnson, Venugopal V. Veeravalli

**Conference on Empirical Methods in Natural Language Processing - EMNLP 2026**



*Towards Context-Based Retrieval in Associative Memories*

Moulik Choraria\*, **Argyrios Gerogiannis**\*, Vidhata Jayaraman, Ankur Mani, Lav R. Varshney

\*Equal contribution

**ICLR 2026 New Frontiers in Associative Memories Workshop (Oral Presentation)**



*Your Model Diversity, Not Method, Determines Reasoning Strategy*

M Choraria, **A Gerogiannis**, A Das, S Chakraborty, B Kapusuzoglu, C-H Lee, K Balasubramaniam, S-X Zhang, S Sahu

**ICLR 2026 Workshop on Logical Reasoning of Large Language Models**



*Learning Where to Look: UCB-Driven Controlled Sensing for Quickest Change Detection*

Yu-Han Huang, **Argyrios Gerogiannis**, Subhonmesh Bose, Venugopal V. Veeravalli

**IEEE Conference on Decision and Control - CDC 2026 (Oral Presentation)**



Detection Augmented Bandit Procedures for Piecewise Stationary MABs: A Modular Approach  
Yu-Han Huang, **Argyrios Gerogiannis**, Subhonmesh Bose, Venugopal V. Veeravalli 

**IEEE Transactions on Information Theory - TIT 2026**

Track-MDP: Reinforcement Learning for Target Tracking with Controlled Sensing (Extended Version) 

Adarsh M. Subramaniam, **Argyrios Gerogiannis**, James Z. Hare, Venugopal V. Veeravalli

**IEEE Transactions on Signal Processing - TSP 2026**

Is Prior-Free Black-Box Non-Stationary Reinforcement Learning Feasible? 

**Argyrios Gerogiannis**, Yu-Han Huang, Venugopal V. Veeravalli

**International Conference on Artificial Intelligence and Statistics - AISTATS 2025**

Track-MDP++: RL for Target Tracking with General Controlled Sensing Models 

Adarsh M. Subramaniam, **Argyrios Gerogiannis**, James Z. Hare, Venugopal V. Veeravalli

**IEEE Military Communications Conference - MILCOM 2025**

Track-MDP: Reinforcement Learning for Target Tracking with Controlled Sensing 

Adarsh M. Subramaniam, **Argyrios Gerogiannis**, James Z. Hare, Venugopal V. Veeravalli

**International Conference on Acoustics, Speech, and Signal Processing - ICASSP 2025**

## Manuscripts Under Review

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*SALSA: Speech Aware LLM Adaptation via Learned Steering Activation Vectors* 

Yekaterina Yegorova, **Argyrios Gerogiannis**, Haolong Zheng, Julia Hockenmaier, Chang D. Yoo, Mark Hasegawa-Johnson

**Under Submission, 2026**

*Know Thy Reasoner: Not All Language Models Explore Alike* 

Moulik Choraria, **Argyrios Gerogiannis**, Anirban Das, Supriyo Chakraborty, Sourya Basu, Sambit Sahu, Lav R. Varshney

**Under Submission, 2026**

*Context-Gated Associative Retrieval: From Theory to Transformers* 

Moulik Choraria\*, **Argyrios Gerogiannis**\*, Vidhata Jayaraman, Ankur Mani, Lav R. Varshney

\*Equal contribution

**Under Submission, 2026**

*DARLING: Detection Augmented Reinforcement Learning with Non-Stationary Guarantees* 

**Argyrios Gerogiannis**, Yu-Han Huang, Venugopal V. Veeravalli

**Under Submission, 2026**

## Skills

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|--------------------|----------------------------------------------------------------------------------------------------------|
| <b>Programming</b> | Python, C++, C, MATLAB                                                                                   |
| <b>Frameworks</b>  | PyTorch, transformers, openrlhf, vLLM, RLlib, ray, TensorFlow, NumPy, pandas, scikit-learn, Hugging Face |
| <b>Development</b> | Git, Docker, AWS, Slurm, Weights&Biases, MLflow, comet                                                   |
| <b>Languages</b>   | Greek (native), English (fluent), Spanish (basic)                                                        |

## Services & Teaching

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**Reviewer** NeurIPS (2026), ICML (2026), AAAI (2027), TMLR (2026), ISIT (2024-2026)

**Teaching Assistant** UIUC: ECE 561 (2026), ECE 445 (2026), ECE 490 (2025), ECE 313 (2025), ECE 598GM (2023)

## Honors & Awards

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### Gold Reviewer Award

Awarded to top reviewers for ICML 2026; included complimentary registration valued at \$1,030.

ICML, USA

### Dilip and Sandhya Sarwate Graduate Fellowship

Awarded to outstanding graduate students.

UIUC, USA

### University Student Excellence Award

Awarded to the student with the highest GPA among all graduates of the ECE department.

UoP, Greece

### Student Excellence Award

Awarded to the student with the highest GPA among all graduates of each department.

IKY, Greece

### Freshman Excellence Award

Awarded to the top 3 freshman students with the highest grade in the entrance exams of the ECE department.

UoP, Greece